Program 6 : Working with Selenium: Maven Project To Connect all the above 3 websites To Web Servers Using Selenium Dependency on Mozilla Firefox

Step 1: create a maven project on terminal by typing the command **mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false**

```
sumanth@sumanth-VirtualBox: ~/devops
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtif
actiId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...chetype-quickstart -DinteractiveMode=false
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp05
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenSeleniumApp05
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.531 s
[INFO] Finished at: 2026-03-19T15:08:59+05:30
[INFO] ------------------------------------------------------------------------
sumanth@sumanth-VirtualBox:~/devops$
```

Step 2 : edit the pom.xml file and add the dependencies

The pom.xml has the code

```
Open   pom.xml  ~/Devops/MyMavenSeleniumApp05   Save
1  <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
2    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
3    <modelVersion>4.0.0</modelVersion>
4    <groupId>com.example</groupId>
5    <artifactId>MyMavenSeleniumApp05</artifactId>
6    <packaging>jar</packaging>
7    <version>1.0-SNAPSHOT</version>
8    <name>MyMavenSeleniumApp05</name>
9    <url>http://maven.apache.org</url>
10   <dependencies>
11     <dependency>
12       <groupId>junit</groupId>
13       <artifactId>junit</artifactId>
14       <version>3.8.1</version>
15       <scope>test</scope>
16     </dependency>
17     <dependency>
18     <groupId>org.seleniumhq.selenium</groupId>
19     <artifactId>selenium-java</artifactId>
20     <version>4.41.0</version>
21     <scope>compile</scope>
22 </dependency>
23   </dependencies>
24   <!-- Build: Configuring plugins and build settings -->
25 <build>
26 <plugins>
27 <!-- Example: Maven Compiler Plugin to compile Java code -->
28 <plugin>
29 <groupId>org.apache.maven.plugins</groupId>
30 <artifactId>maven-compiler-plugin</artifactId>
31 <version>3.8.1</version>
32 <configuration>
33 <source>1.7</source>
34 <target>1.7</target>
35 </configuration>
36 </plugin>
```

pom.xml
~/Devops/MyMavenSeleniumApp05

```xml
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>


        </plugin>
        <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-shade-plugin</artifactId>
      <version>3.5.0</version>
      <executions>
        <execution>
          <phase>package</phase>
          <goals>
            <goal>shade</goal>
          </goals>
          <configuration>
            <transformers>
              <transformer implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">
                <mainClass>com.example.App</mainClass>
              </transformer>
            </transformers>
          </configuration>
        </execution>
      </executions>
    </plugin>

</plugins>
</build>
</project>
```

Step 3: **<u>Edit the App.java</u>**

*App.java
~/Devops/MyMavenSeleniumApp05/src/main/java/com/example

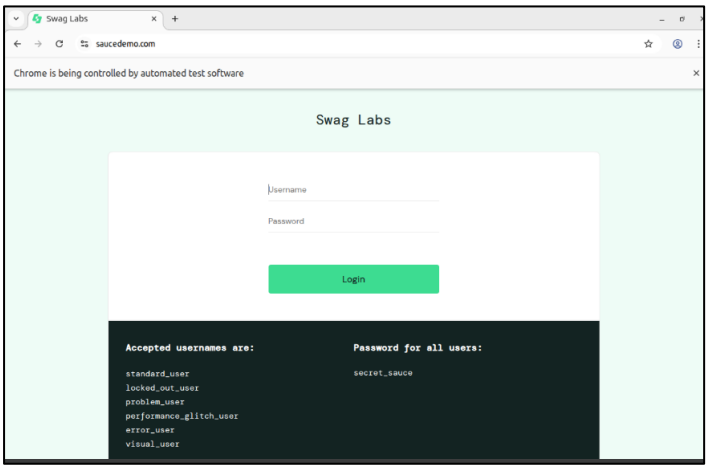
```java
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);

        driver.get("https://www.saucedemo.com/");

        driver.findElement(By.id("user-name")).sendKeys("standard_user");

        driver.findElement(By.id("password")).sendKeys("secret_sauce");

        driver.findElement(By.id("login-button")).click();

        System.out.println("SauceDemo login successful");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://automationexercise.com/products");
```

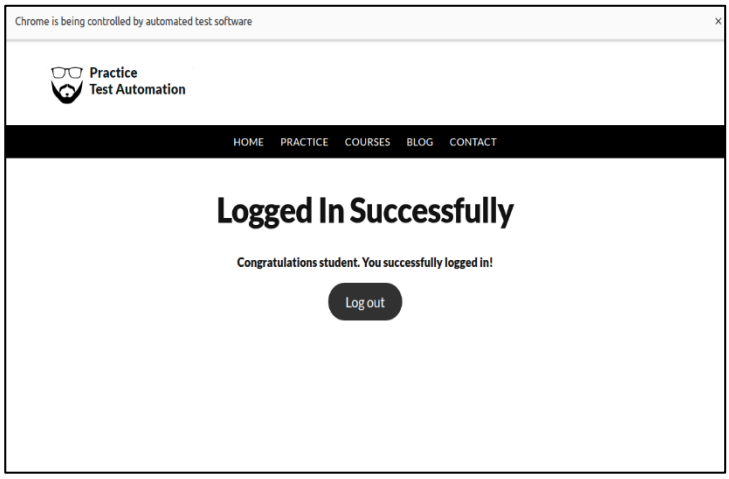
```java
        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView({block:'center'});", product);

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", product);

        WebElement viewCart = wait.until(
                ExpectedConditions.elementToBeClickable(
                        By.cssSelector("#cartModal a[href='/view_cart']")
                )
        );

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].scrollIntoView(true);", viewCart);

        ((org.openqa.selenium.JavascriptExecutor) driver)
                .executeScript("arguments[0].click();", viewCart);

        System.out.println("Automation Exercise product added to cart");

        driver.switchTo().newWindow(WindowType.TAB);

        driver.get("https://practicetestautomation.com/practice-test-login/");

        Thread.sleep(2000);

        driver.findElement(By.id("username")).sendKeys("student");

        driver.findElement(By.id("password")).sendKeys("Password123");

        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
}
```

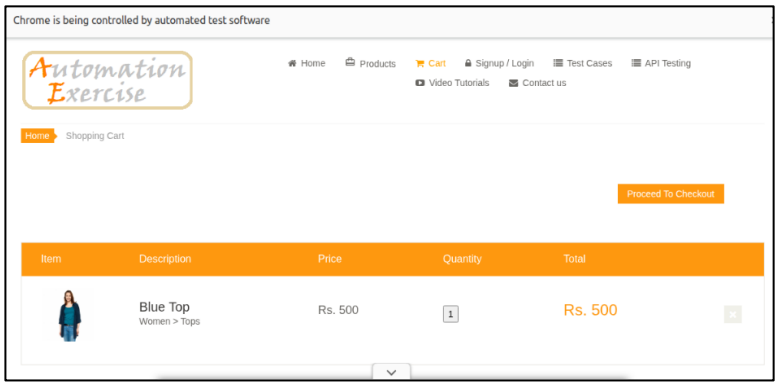
Step 4: build and run the maven application

To execute type the command **mvn exec:java -Dexec.mainClass="com.example.App"**

it opens all the 3 websites automatically and registers using username and password.

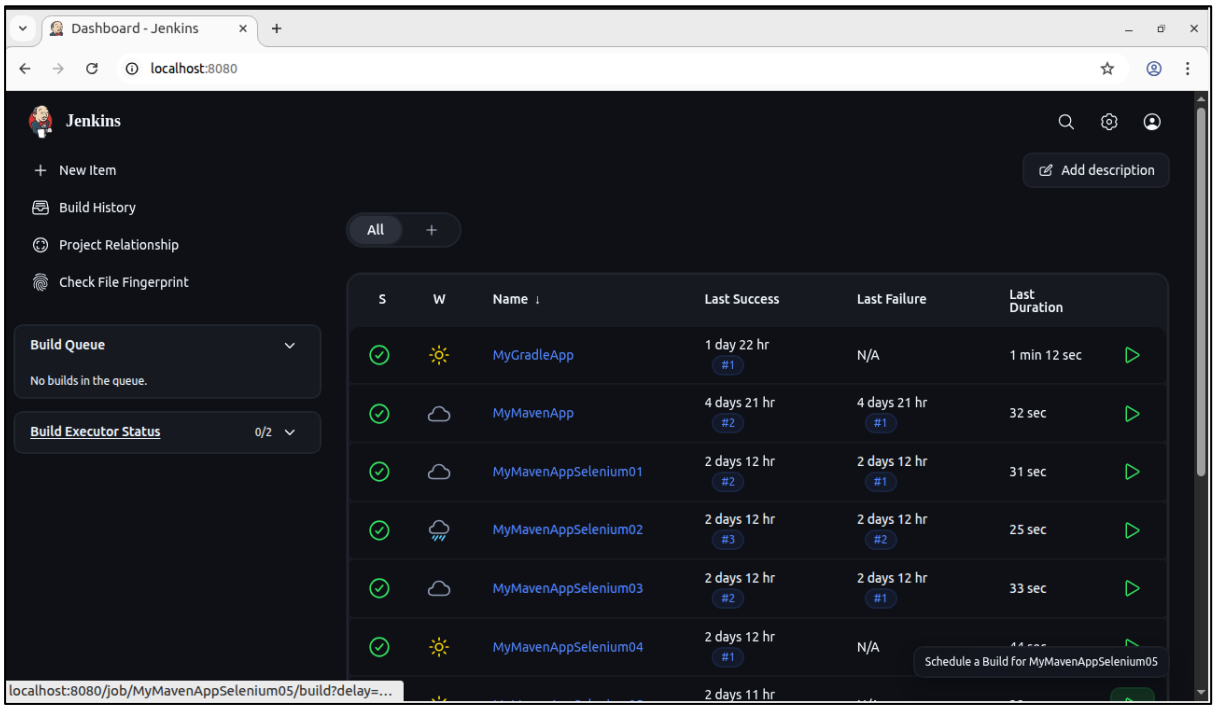






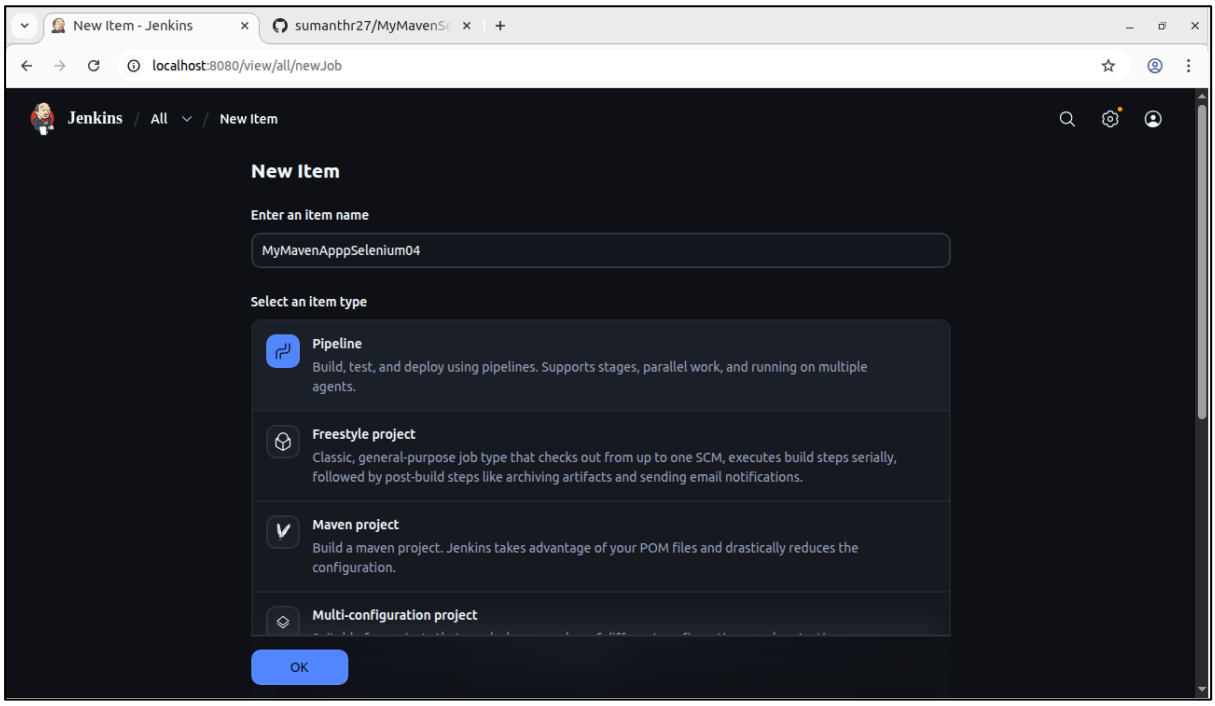
Jenkins Pipeline for MyMavenSelenium04

Step 1: Open the Jenkins Dashboard in order to create a new item

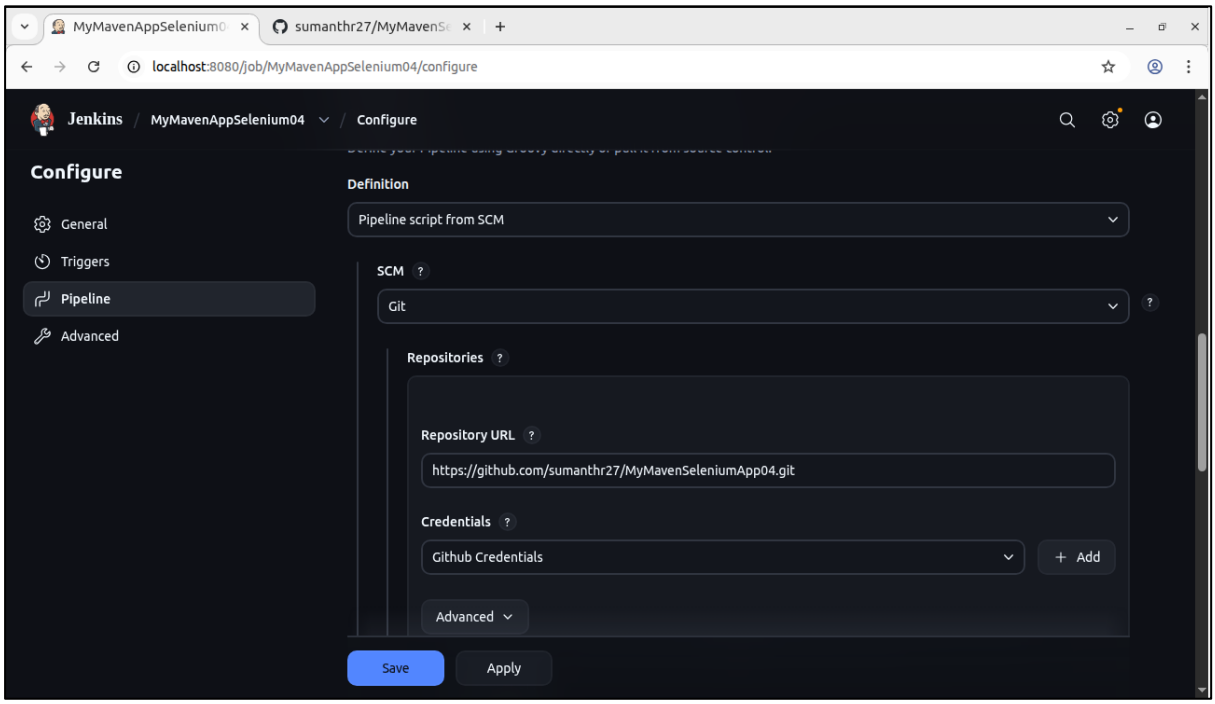


Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenAppSelenium04” and Item type as Pipeline and

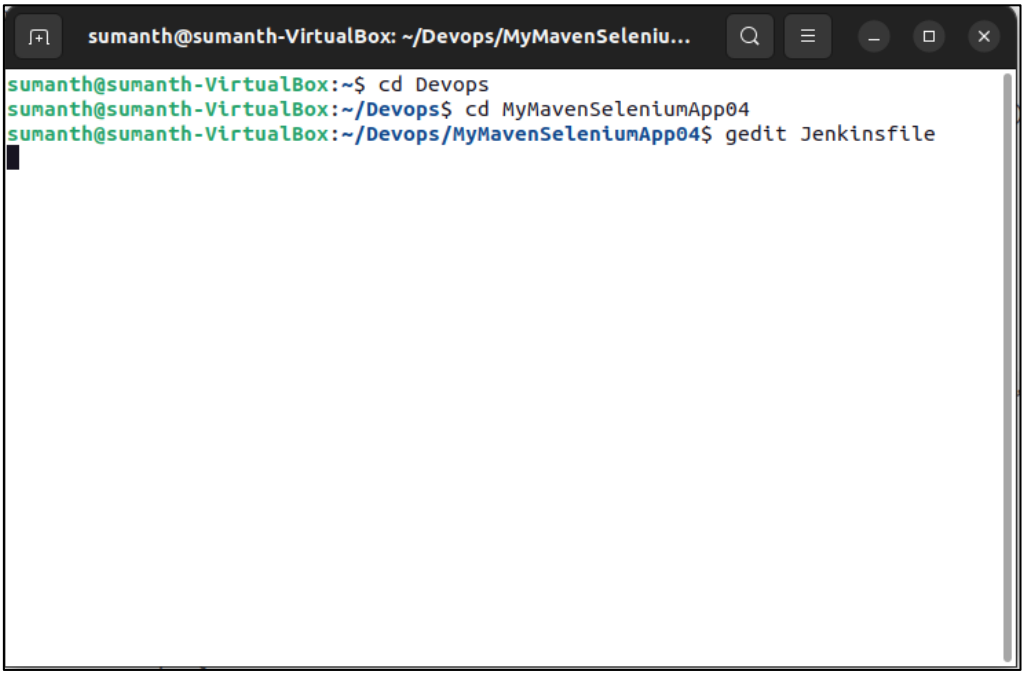
Create it



Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenAppSelenium03 and create an new file called “Jenkinsfile”



Step 5: Paste the available code inside the Jenkinsfile

Jenkinsfile
~/Devops/MyMavenSeleniumApp04

```
pipeline {
    agent any   // Run on any available agent

    tools {
        maven 'Maven'   // Must match name configured in Jenkins (Global Tool Config)
    }

    stages {

        stage('Checkout') {
            steps {
                // Recommended if repo is configured in Jenkins
                checkout scm

                // OR use this if not configured:
                // git url: 'https://github.com/sumanthr27/MyMavenSeleniumApp04.git', branch: 'master'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }

    post {
        always {
            echo 'Pipeline execution completed'
        }
        success {
```

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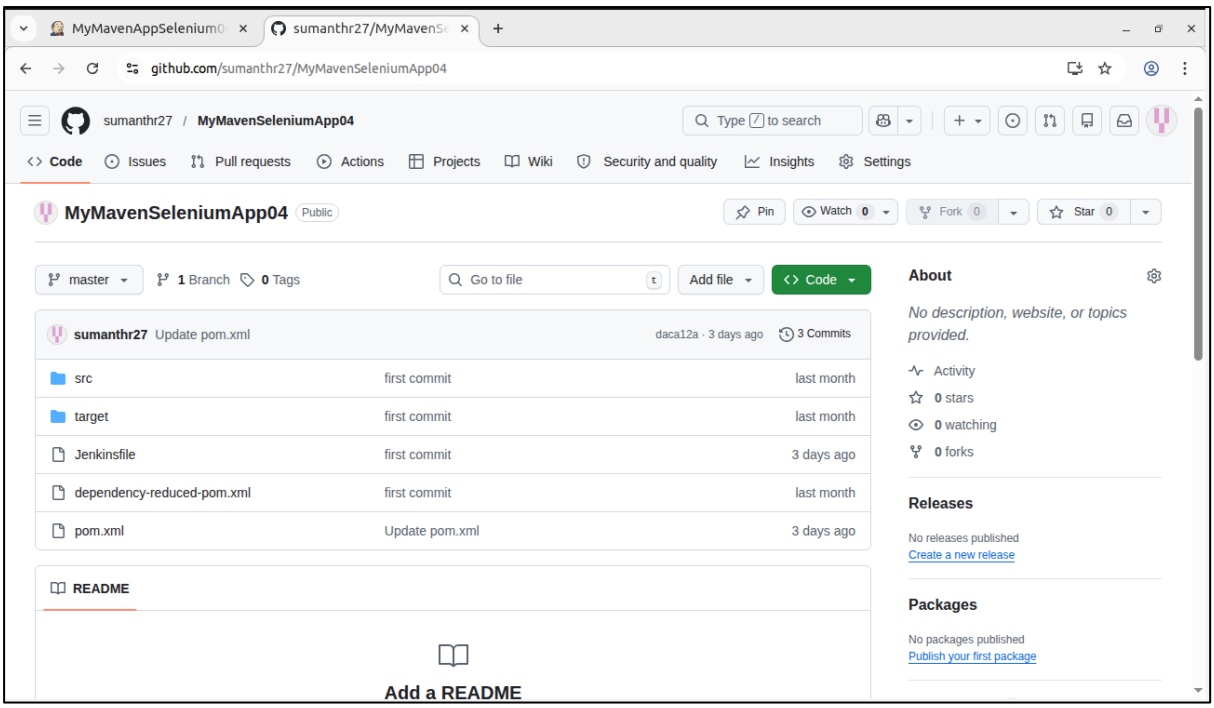
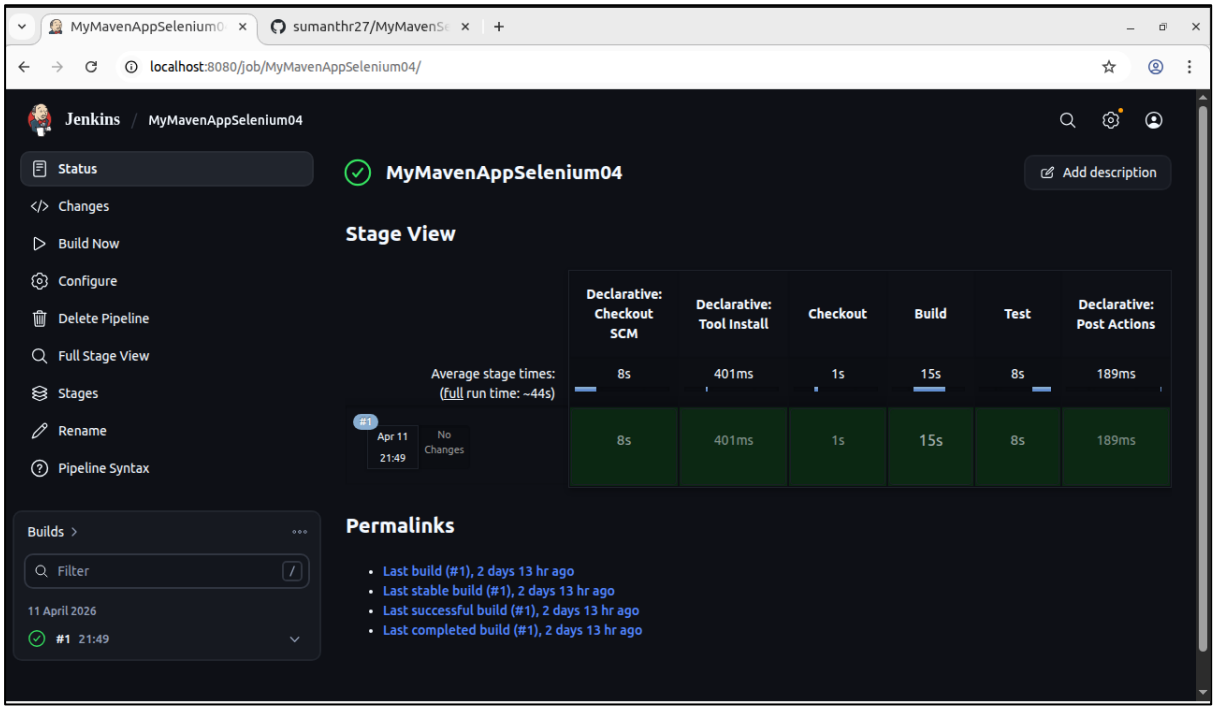
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.